AI.308

Intelligent Control Syst

COURSE OVERVIEW

Modern control: adaptive, robust, and intelligent control for uncertain robotic plants.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: BEYOND PID

KEY TOPICS

- Limitations of classical control
- Intelligent control
- Adaptive and robust control
- Applications

TEACHING FOCUS

Control systems that adapt to change.

KEY TAKEAWAY Intelligent control handles uncertainty.

2

SESSION

60 MINUTES

ADAPTIVE CONTROL

KEY TOPICS

- Model-reference adaptive control
- Self-tuning regulators
- Parameter estimation
- Gain scheduling

TEACHING FOCUS

Adjust the controller as the plant changes.

KEY TAKEAWAY Adaptation keeps performance as plants drift.

3

SESSION

60 MINUTES

ROBUST CONTROL

KEY TOPICS

- H-infinity and mu-synthesis
- Uncertainty modeling
- Robust stability
- Loop shaping

TEACHING FOCUS

Guarantee performance under uncertainty.

KEY TAKEAWAY Robustness is control for the unknown.

4

SESSION

60 MINUTES

INTELLIGENT CONTROL METHODS

KEY TOPICS

- Fuzzy control
- Neural control
- Genetic-algorithm-based tuning
- Hybrid intelligent control

TEACHING FOCUS

Combine learning and control.

KEY TAKEAWAY Learning + control = intelligent control.

5

SESSION

60 MINUTES

CONTROL FOR ROBOTICS

KEY TOPICS

- Impedance and admittance control
- Force control
- Whole-body control
- Physical human–robot interaction

TEACHING FOCUS

Control robots that touch the world.

KEY TAKEAWAY Intelligent control enables safe contact.

6

SESSION

60 MINUTES

INTELLIGENT CONTROL & REVIEW

KEY TOPICS

- Controller implementation
- Course recap and assessment
- Toward multi-robot systems (AI.309)
- Control lab

TEACHING FOCUS

Tune an adaptive controller on hardware.

KEY TAKEAWAY The best controllers learn as they act.